CRISPR in action: practice and theory of genome engineering

Day 5: Analyzing genotyping results

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Course overview - experiments

Day

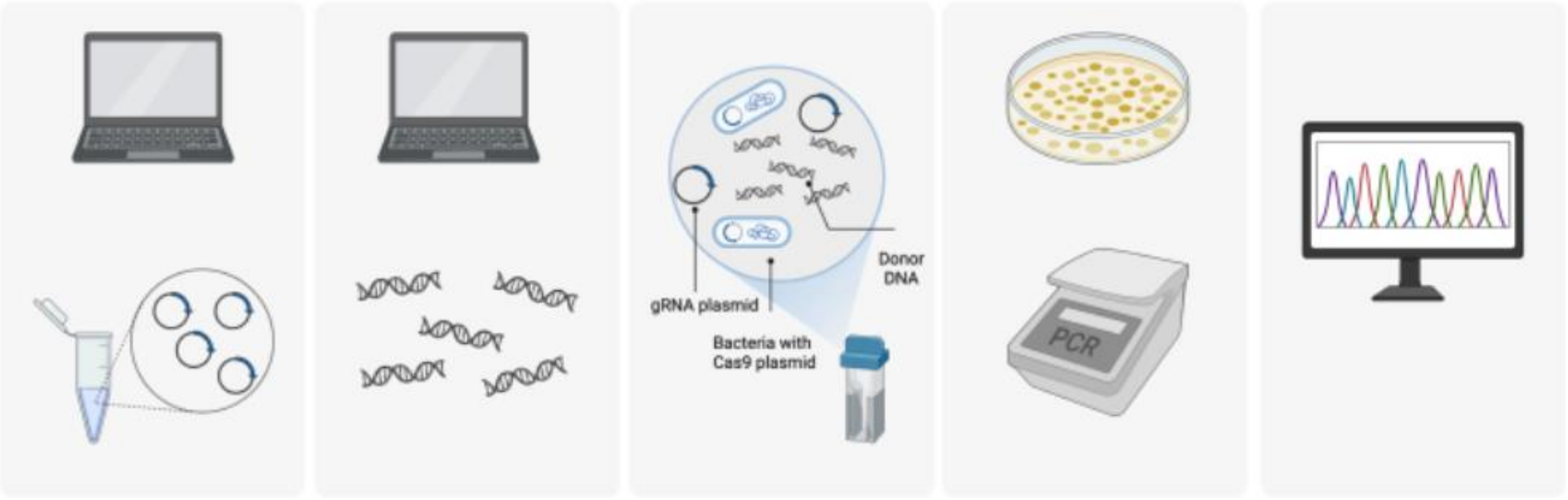
1
- gRNA design
- gRNA cloning

2
- donor design
- donor assembly

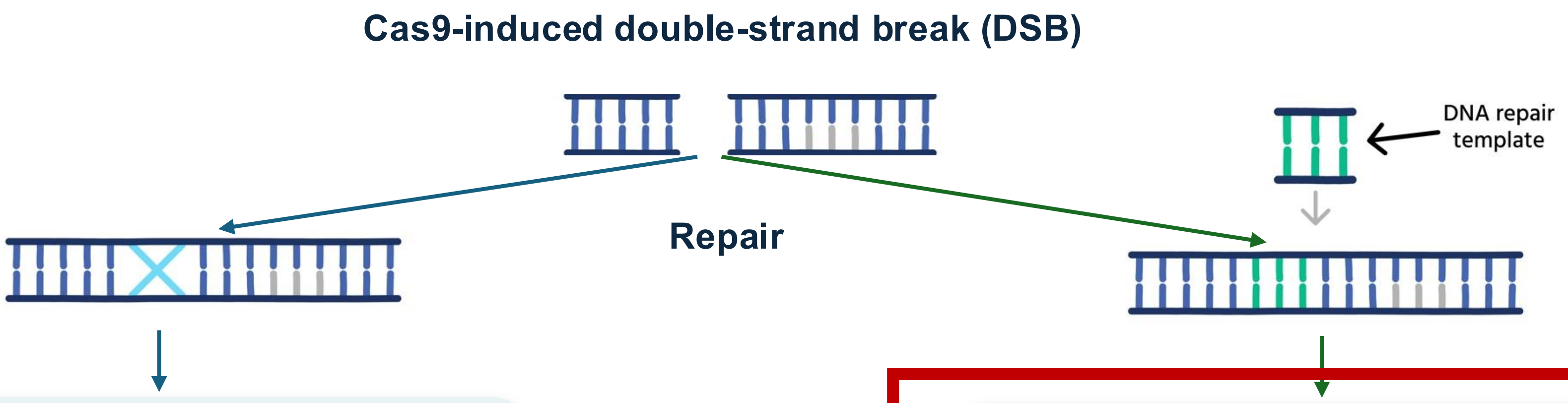
3
- editing

4
- genotyping

5
- analysis



Recap on NHEJ and HR



NHEJ

Non-Homologous End Joining

- Directly re-ligates the two broken ends
- Active throughout the cell cycle — the dominant pathway in most cells
- Error-prone: often leaves small insertions/deletions (indels)

→ Gene knockout / disruption

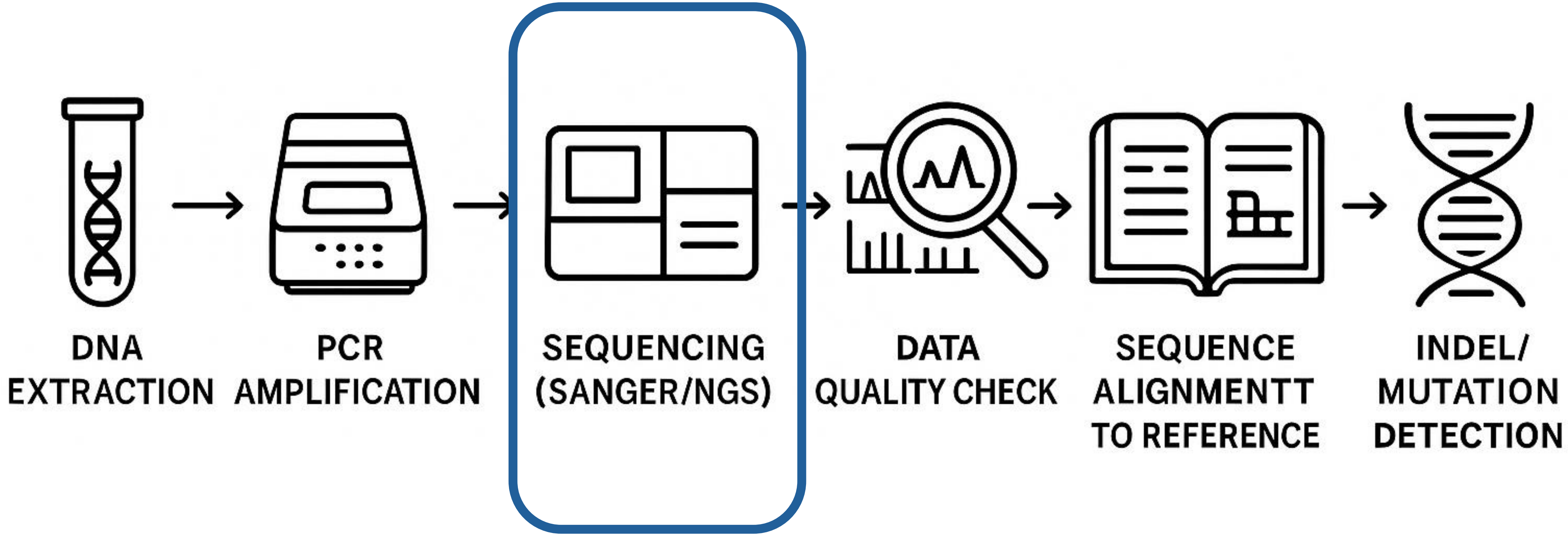
HDR

Homology-Directed Repair

- Copies across the break using a homologous template (sister chromatid or donor DNA)
- Restricted to S/G2 phase — naturally less frequent
- High-fidelity: enables precise edits, insertions or corrections

→ Precise edit / knock-in

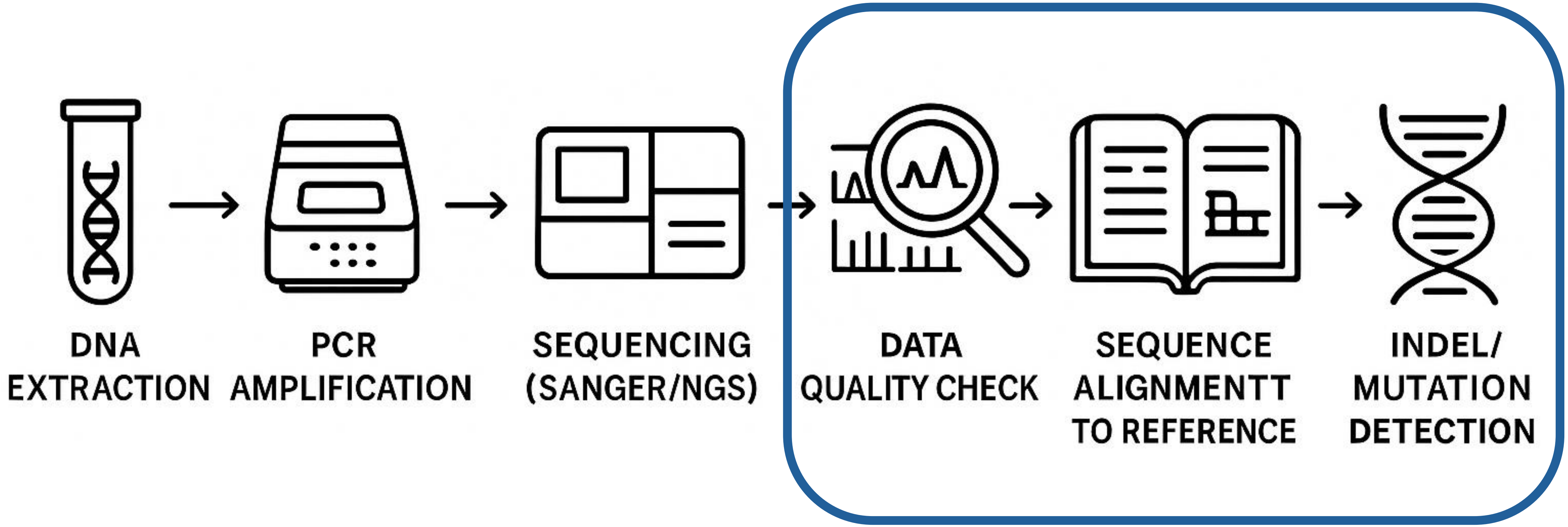
Genotyping analysis



Sanger vs. NGS vs. Nanopore

Feature	Sanger Sequencing	Next-Generation Sequencing (NGS)	Nanopore Sequencing
Read Length	Long reads (~700-1000 bp)	Short reads (~50-300 bp)	Ultra-long reads (up to megabases)
Throughput	Low (single/few sequences per run)	High (millions of reads/run)	Moderate-high, real-time reads
Turnaround Time	1 day for a few targets	Depends on access to a sequencing facility	Near real-time possible
Cost per Sample	High for many samples, good for single targets	Low for large-scale, high for few samples	Moderate, scalable (device/protocol-dependent)
Data Output	Chromatograms, easy to interpret (Benchling)	Complex datasets, bioinformatics pipeline needed	Bioinformatics pipeline needed
Sensitivity	Detects major variants, limited for mixtures	Highly sensitive, detects low-frequency edits	Can resolve structural/complex variants.
Application Scale	Single/locus validation, clone analysis	Large scale, screening, diversity analysis	Structural variants, haplotyping, metagenomics
Error Rate per Nucleotide	~0.1% (1 in 1000 bases)	~0.1%–1% (depends on platform; Illumina ~0.1%)	~5%–10% (improving, varies by chemistry/software)
Use Case Example	Confirming a CRISPR edit in clone	Screening pools, high-throughput variant calling	Rare/large variant analysis

Genotyping analysis



Overview of software tools and pipelines for NGS data

- **Geneious Prime/QIAGEN CLC Main Workbench:** Graphical software for both Sanger and NGS, for visualizations, and guided analysis workflows.
- **CRISPResso:** Web and command-line, widely used for in-depth analysis of editing outcomes (insertions, deletions, substitutions, quantification).

1. Command line version available
https://github.com/pinellolab/CRISPResso2

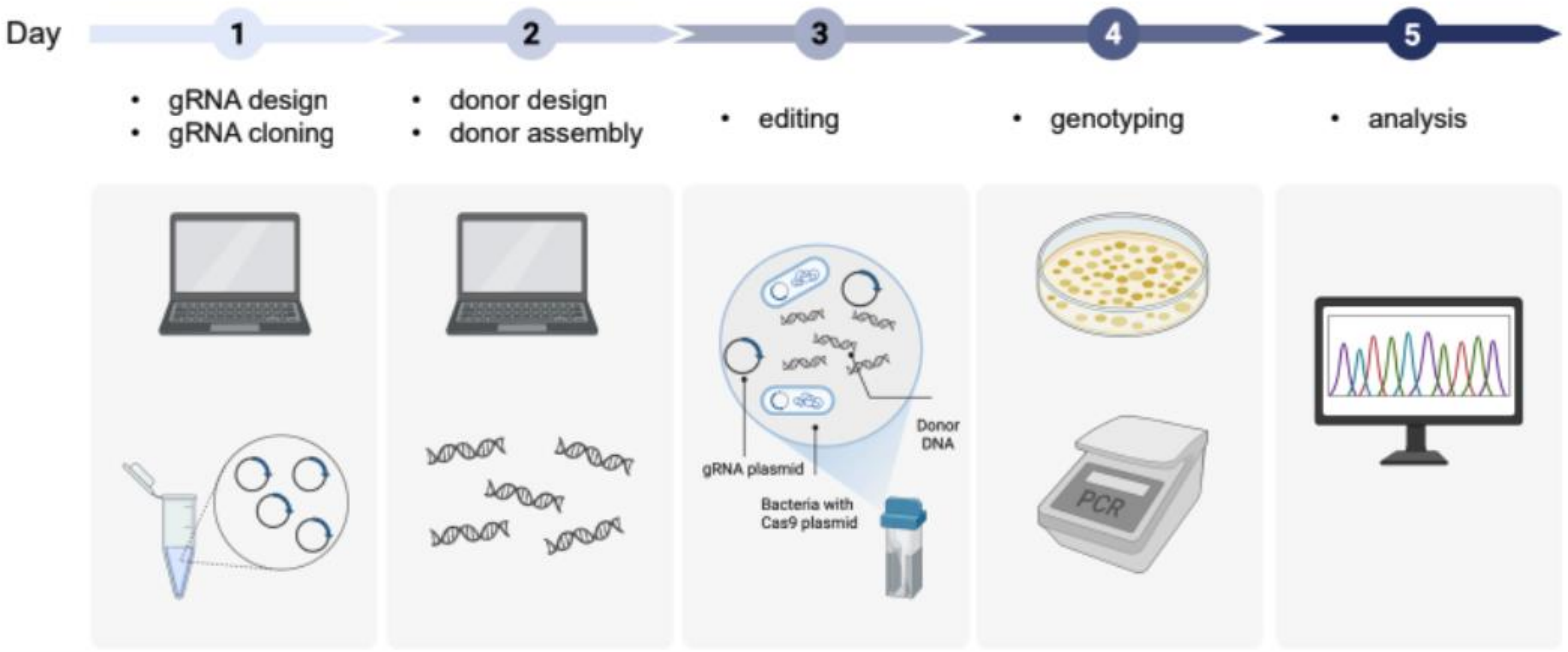
2. Web interface (no-code version)
https://crispresso2.pinellolab.org

CRISPResso2

Analysis of genome editing outcomes from deep sequencing data

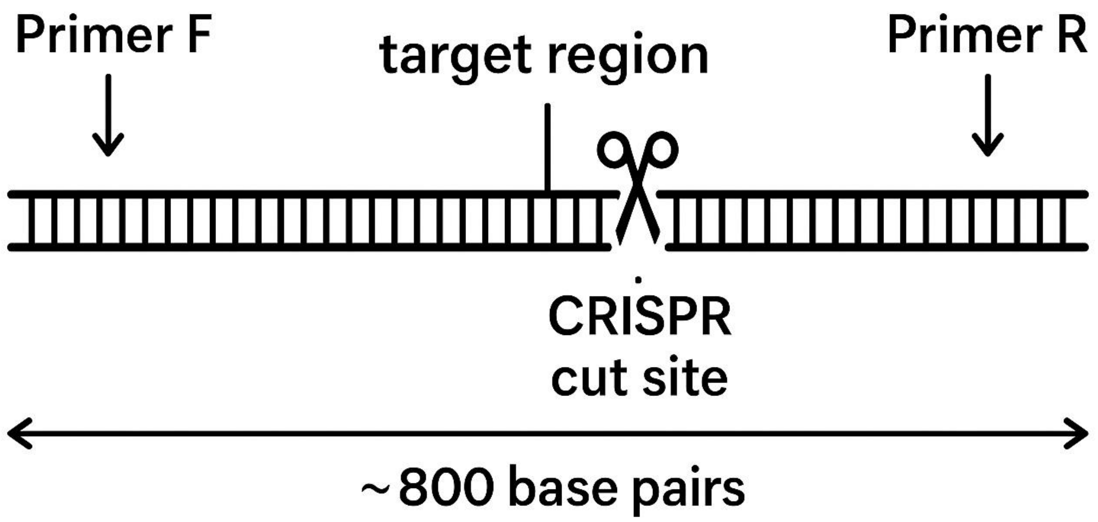
Clement et al. 2019
https://doi.org/10.1038/s41587-019-0032-3

Course overview - experiments



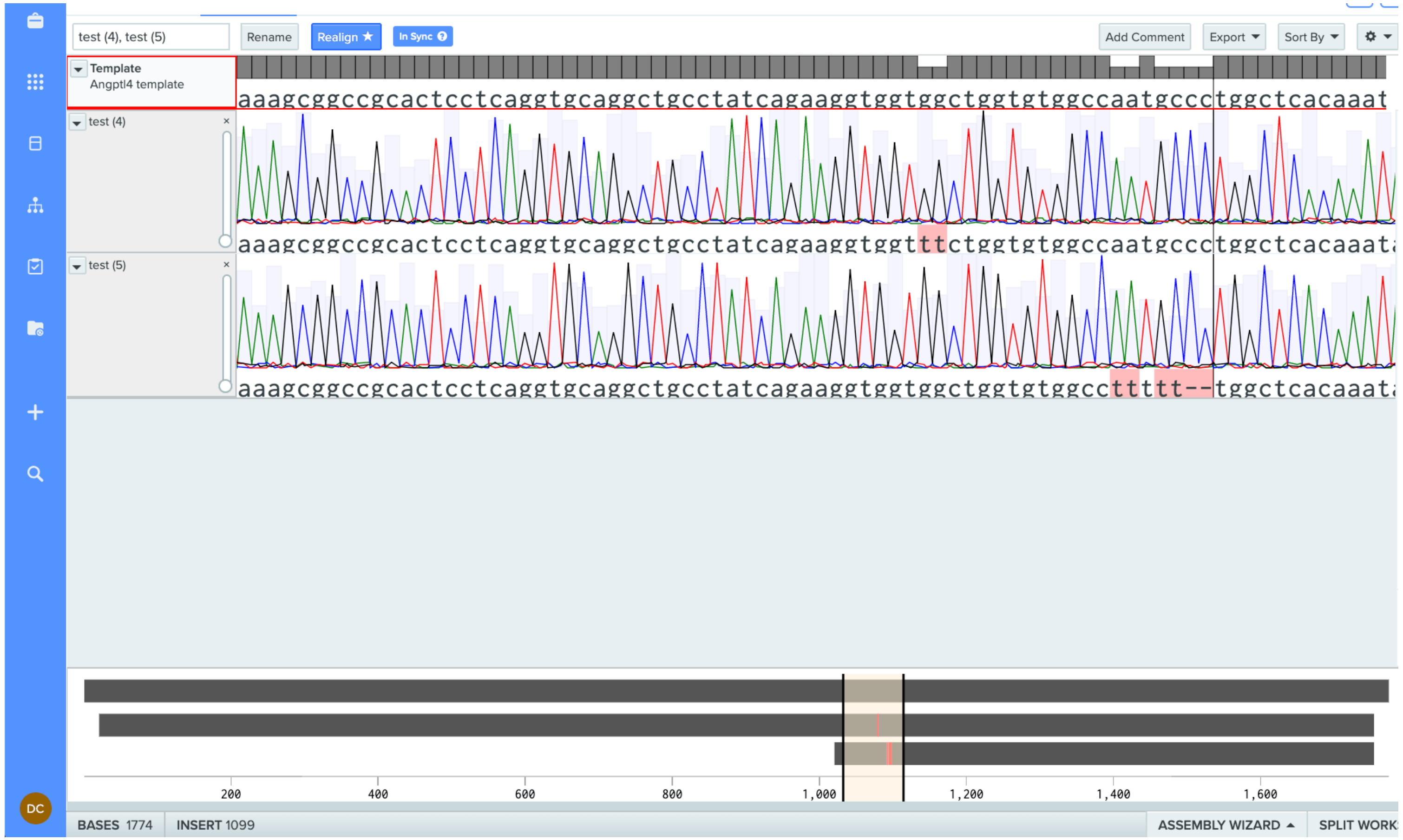
How amplicons are (Sanger) sequenced to span the CRISPR cleavage site

- Amplicon (= PCR product)
- Amplicon should cover both the gRNA binding site and the expected edit location
- Fragment should be more than 200 bp long, ideally between 800-1000 bps. The amplicon must be long enough to capture edits
- The region of interest where the gRNA will cut should be approx. 150 bp away from the primer you will use for sequencing. The quality close the the primer will be too low



Typical analysis workflow for Sanger

1. **Data quality assesment.** Checking chromatograms.
2. **Alignment:** Placing reads against the reference to spot mutations (Benchling)
3. **Detection of indels and variants:** Visualizing and quantifying edits, including estimation of allele frequencies.
4. **Interpreting biological outcomes:** Link sequence variants to possible phenotypic effects.



- High of the peak reflects signal intensity, not sequencing quality
- Very high or very low peaks might sometimes indicate technical problems, but differences in peak height do not always signify errors. Consistency and clarity are more important than uniform height across the chromatogram
- Heterozygous or mixed bases can show as double or overlapping peaks: In such cases, peak height ratios can give information about allele balance

Overview of software tools for Sanger

- **TIDE:** https://tide.nki.nl/
- **DECODR:** Accepts Sanger data and deconvolutes mixed traces, user-friendly online platform (https://decodr.org/)
- **ICE:** https://www.synthego.com/products/bioinformatics/analysis
- **Benchling:** importing, aligning, and annotating Sanger sequencing data (including consensus sequence generation and mutation confirmation for CRISPR edits), but does not automate deconvolution of mixed traces or advanced CRISPR editing quantification.

From standard **Sanger sequencing**, you typically get a few different types of files from the sequencing facility:

- **.ab1 files** (Applied Biosystems chromatogram files)
  - These are the raw output files generated by the sequencer.
  - They contain the chromatogram (trace data: the colored peaks representing A, T, G, C), the base calls, and quality scores.
  - You can open and inspect them in programs like **Benchling, SnapGene or Geneious**.
- **.seq or .fasta files** (plain text sequence files)
  - These are extracted base calls only, without the chromatogram.
  - Sometimes provided by the facility as a convenience.
- **.txt files** (optional)
  - Some facilities provide the sequence as plain text in addition to the raw .ab1 file.

Usually, you'll at least get **.ab1 files** (raw traces), and often also **.seq/.fasta files** (cleaned sequences).